



spca: an R package to compute Least Squares Sparse Principal Components

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Abstract

This article illustrates the use of the package *spca* how to write a manuscript for the i.e., avoid inclusion of packages/commands that are not necessary. For outlining the typical structure of a JSS article some brief text snippets are employed that have been inspired by discussing count data regression in R. Editorial comments and instructions are marked by vertical bars.

Keywords: JSS, SPCA, Least squares, LS, uncorrelated.

1. Introduction to LS-SPCA

sparse principal components analysis (SPCA) is a variant of principal components analysis (PCA) that aims at computing sparse principal components (sPCs) that are linear combinations of only a few of the features, hence having *sparse* loadings with few nonzero elements.

Most of the existing *conventional* SPCA methods compute sPCs that have maximal variance, using the objective function defined by Hotelling (1933). There exist a plethora of such methods which, as outlined in the *All sparse PCA models are wrong, but some are useful* series of review articles (Camacho, Smilde, Saccenti, and Westerhuis, 2020, 2021, 2025), produce competing solutions, that are often not close to the original principal components (PCs). Merola and Chen (2019) also shows other drawbacks of these methods.

LS-SPCA was first proposed in Merola (2015) and its projection variant in Merola and Chen (2019). This approach aims to compute sPCs that can be used to best approximate the data, in a least-squares (LS) sense, hence adding sparsity constraints to (Pearson 1901) classic PCA's data approximation objective. It should be noticed that, while Pearson's and Hotelling's objectives give the same solutions in PCA, this is not the case when sparsity constraints are introduced (Merola 2015; Merola and Chen 2019), and this is probably the

reason why [Camacho et al.](#) found that all (conventional) SPCA methods are wrong.

The **spca** package computes LS-SPCA solutions with several options and helper functions to visualize and compare the solutions. Rather than a black-box optimizer type function, the package provides an intuitive UI with transparent options and goodness-of-fit metrics.

This article will be used as extended vignettes in the package, so will illustrate several of the available options and helpers, useful to conduct a thorough SPCA. In the next Section we briefly introduce the theory behind LS-SPCA. In Section ?? we illustrate the package with examples.

1.1. LS-SPCA estimation

The material in this section is taken from [Merola \(2015\)](#) and [Merola and Chen \(2019\)](#) where proofs and further details can be found. Let's establish notation first. X denotes the $n \times p$ data matrix containing n cases on p features centered to mean zero. $S \propto X^\top X$ denotes the covariance matrix. The eigenvectors of S , $v_1, \dots, v_p$ ($\|v_j\| = 1$, $j = 1, \dots, p$ (the PCs loadings)) are ordered by decreasing corresponding eigenvalues, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$. The same convention is used for all eigenvectors (including generalized ones).

The sparse loadings are the vectors $a_j = (a_{i1}, \dots, a_{ir}, r \leq p$ with unit norm $a_j^\top a_j = 1$, each of which has only $c_j \leq p$ non-zero elements. The quantity c_j is called the *cardinality* of a_j . The sPCs are then equal to $t_j = Xa_j$.

A dot above a symbol denote the operation of retaining the elements of a vector or the columns of a matrix corresponding to a set of indices, ind_j . In R syntax, we write $\dot{a}_j = a_j[ind_j]$ to indicate the vector containing only the c_j non-zero loadings in $\dot{a}_j$, so that $t_j = \dot{X}_j \dot{a}_j$.

There exist there variants of LS-SPCA, optimizing two different objectives LS-SPCA solutions are derived using two different objectives.

Objective: maximal Variance explained

The SPCA objective obtained by adding sparsity constraints to the PCA objective is

$$\min_{a_j} \|X - t_j b^\top\|^2, \quad \text{subject to} \quad \text{card}(a_j) < p, \quad a_j^\top S a_i = 0, \quad i < j; j = 1, \dots, r. \quad (1)$$

Uncorrelated LS-SPCA (uSPCA). The optimal sparse loadings for objective 1, $\dot{a}_j$, are the generalized eigenvectors corresponding to the largest eigenvalue, λ_j , solution to

$$C_j \dot{S}_j^\top \dot{S}_j \dot{a}_j = \lambda_j \dot{X}_j^\top \dot{X}_j \dot{a}_j, \quad (2)$$

where the matrices C_j ($C_1 = I$) enforce the orthogonality constraints.

The orthogonality constraints require that the cardinality of the solution loadings satisfy $c_j \geq j$. Furthermore, the solutions can be computationally demanding and unstable.

Correlated LS-SPCA (cSPCA). Approximated but simpler to compute solutions can be obtained from the residual matrices orthogonal to the previously computed sPCs. Let

$$Q_j = \left(I_p - \frac{t_{j-1} t_{j-1}^\top}{t_{j-1}^\top t_{j-1}} \right) Q_{j-1} = H_j Q_{j-1}, \quad j = 2, 3, \dots, r, \quad (3)$$

where H_j is the *hat* matrix and $Q_1 = X$. IN SPCA context, the residual matrices Q_j are often called *deflated* matrices. Note that the use of the deflated matrices is a direct consequence of the Courant–Fischer min–max theorem and is typically used in the power method for computing the eigen-decomposition. Then, the cSPCA solution loadings are obtained as the generalized eigenvectors corresponding to the largest eigenvalue:

$$\dot{Q}_j^\top X X^\top \dot{Q}_j \dot{\mathbf{a}}_j = \mu_j \dot{X}_j^\top \dot{X}_j \dot{\mathbf{a}}_j. \quad (4)$$

The first cSPCA solution is equal to the first uSPCA solution but, in general, the following solutions are not equal to the corresponding uSPCA solutions. The cSPCA solutions typically show very mild correlation with one another, with increasing values for sPCs of higher order. However, their cardinality is not constrained by the uncorrelatedness requirements and, in most cases, it is lower than their order.

Maximal correlation with the PCs (pSPCA)

This objective function requires that the sPCs approximate well the PCs. Since the PCs best approximate the data, the resulting sPCs will be close to the ones computed with the other objective, and vice versa, However, this approach is computationally simpler.

The objective function is

$$\max_{a_j} |\text{cor}(t_j, y_j)|, \quad \text{subject to } \text{card}(a_j) < p. \quad (5)$$

It is well known that the optimal solution to this objective is linear regression. Since starting with the first, the sPCs explain less variance than the corresponding PCs, using the first PC of the Q_j matrices, instead of the PCs, allows to recover some of this unexplained variance.

Therefore, in pSPCA, the sPCs are obtained by projecting the residual PCs onto subsets of the x -variables. Let $z_j = Q_j v_j$ denote the first PC of the deflated matrix Q_j . Then the pSPCA solution loadings are obtained as the regression coefficients, standardized to unit norm, from the regression of z_j onto $\dot{X}_j$:

$$\dot{\mathbf{a}}_j \propto (\dot{X}_j^\top \dot{X}_j)^{-1} \dot{X}_j^\top z_j. \quad (6)$$

The psPCs have maximum multiple squared correlation (the R^2 coefficient) for the given set of indices ind_j .

The sPCs obtained by maximizing the variance explained are usually highly correlated with the corresponding PCs. However, as the order of the components increases, these correlations tend to decrease.

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This objective function requires that the sPCs approximate well the PCs. Since the PCs best approximate the data, the resulting sPCs will be close to the ones computed with the other objective, and vice versa, However, this approach is computationally simpler.

$$\max_{a_j} |\text{cor}(t_j, y_j)|, \quad \text{subject to } a_j^\top S a_i = 0, \quad i < j, \quad \text{card}(a_j) < p. \quad (7)$$

1.2. LS-SPCA solutions

We now face two challenges. The first is to derive the solutions when the indices of the non-zero loadings are known. The second is to identify the optimal set of these indices for each sPC.

We begin by assuming that these indices are known. Let ind_j denote the set of indices for the j th sPC, and let a dot above a symbol denote the operation of retaining the elements of a vector or the columns of a matrix corresponding to these indices. In R syntax, we can write $\dot{a}_j = a_j[\text{ind}_j]$ and $\dot{S}_j = S[, \text{ind}_j]$ and $t_j = \dot{X}\dot{a}_j$.

Maximal variance explained

It can be shown that the solution requires solving

$$\max_{a_j} \frac{a_j^\top X^\top X X^\top X a_j}{a_j^\top X^\top X a_j} \equiv \max_{a_j} \frac{a_j^\top S S a_j}{a_j^\top S a_j}, \quad a_j^\top S a_i = 0, \quad i < j, \quad \text{card}(a_j) < p. \quad (8)$$

The objective function 8 is the *variance explained* (VEXP) by the j th component. Without cardinality constraints it is the objective of standard PCA.

In passing, we note that, without cardinality constraints, objective function 8 is original PCA objective formulated by [Pearson \(1901\)](#), which has solutions

$$\max_{v_j} a_j^\top S a_j, \quad a_j^\top S a_i = 0, \quad i < j.$$

These solutions are [Hotelling \(1933\)](#) definition of PCA objective. The equivalence between the two objectives is no longer valid when cardinality constraints are introduced ([Merola 2015](#)).

Uncorrelated LS-SPCA (uSPCA). The exact solutions to objective 8 are computed by adding the constraints directly into the variance explained expression by means of the generic $\dot{X}$ matrix. Hence the optimal sparse loadings, $\dot{\mathbf{a}}_j$, are the generalized eigenvectors corresponding to the largest eigenvalue, λ_j , solution to

$$C_j \dot{S}_j^\top \dot{S}_j \dot{\mathbf{a}}_j = \lambda_j \dot{X}_j^\top \dot{X}_j \dot{\mathbf{a}}_j, \quad (9)$$

where the matrices C_j ($C_1 = I$) enforce the orthogonality constraints. The orthogonality constraints require that the cardinality of the solution loadings satisfy $c_j \geq j$. Furthermore, the solutions can be computationally demanding and unstable.

Correlated LS-SPCA (cSPCA). Approximate but simpler to compute solutions can be obtained from the orthogonal residuals of the previously computed sPCs. Let

$$Q_j = \left(I_p - \frac{t_{j-1}t_{j-1}^\top}{t_{j-1}^\top t_{j-1}} \right) Q_{j-1} = H_j Q_{j-1}, \quad j = 2, 3, \dots, r, \quad (10)$$

where H_j is the *hat* matrix and $Q_1 = X$. The matrices Q_j are often called *deflated* matrices. Note that the use of the deflated matrices is a direct consequence of the Courant–Fischer min–max theorem and is typically used in the power method for computing the eigendecomposition. Then, the cSPCA solution loadings are obtained as the generalized eigenvectors corresponding to the largest eigenvalue:

$$\dot{Q}_j^\top X X^\top \dot{Q}_j \dot{\mathbf{a}}_j = \mu_j \dot{X}^\top \dot{X} \dot{\mathbf{a}}_j. \quad (11)$$

The first solution is equal to the first uSPCA solution but, in general, the following ones are not equal to the corresponding uSPCA solutions. The cSPCA solutions typically show very mild correlation with one another, with increasing values for sPCs of higher order. However, their cardinality is not constrained by the uncorrelatedness requirements and, in most cases, it is lower than their order.

$$\mu_j = \max_{\mathbf{a}_j} \frac{\mathbf{a}_j^\top X^\top Q_j Q_j^\top X \mathbf{a}_j}{\mathbf{a}_j^\top X^\top X \mathbf{a}_j}. \quad (12)$$

This is an approximate measure of the variance explained. Because the resulting PCs are correlated, the exact VEXP must be obtained as the difference between the cumulative VEXP (CVEXP) explained by the first j csPCs and that explained by the preceding $(j - 1)$ csPCs:

$$\text{VEXP}(t_j) = \text{trace}\left(X^\top T_j (T_j^\top T_j)^{-1} T_j^\top X\right) - \text{trace}\left(X^\top T_{j-1} (T_{j-1}^\top T_{j-1})^{-1} T_{j-1}^\top X\right), \quad (13)$$

where T_j is the matrix with columns equal to the first j sPCs.

1.3. Variable selection

Variable selection is notoriously a computationally demanding, NP-hard task that requires greedy algorithms for its solution.

Regression-based variable selection algorithms are used to determine the pSPCA solution. Thus, we implemented standard forward, backward, and forward stepwise variable selection algorithms used for regression, with a stopping rule based on the current R^2 . Because the response variable z_j is proportional to an eigenvector of Q_j , the algorithms are simplified using the fact that $z_j^\top X = z_j^\top Q_j = v_j \lambda_j$. These algorithms have approximate complexity $O(p^2)$.

For uSPCA and cSPCA we implemented forward and backward algorithms that terminate after a minimal proportion of the CVEXP of the corresponding number of PCs is reached or, respectively, not passed. These algorithms use the approximate VEXP in Equation ?? to determine the variables to select or eliminate and are computationally very expensive. The forward selection is approximately $O(pk^4)$ for selecting k variables, even when VEXP is computed with the power method to save computation time. However, a dramatic reduction in computational complexity can be achieved by selecting the variables using one of the algorithms implemented for pSPCA and computing the CVEXP at each iteration, which is then used as the stopping rule. While the exact cSPCA-based algorithm may yield more efficient solutions, in many cases there is little difference from the results obtained using the regression variable selection algorithms.

2. Computational details

Since uSPCA and cSPCA require solving generalized eigenvector problems, the computation for large matrices is computationally demanding. The problem is even more taxing in variable selection. Therefore, we put some care to make the computation feasible on large matrices.

The major expedient that we use is to do the demanding $O(np^2)$ multiplication to compute the covariance matrix $S = X^\top X$ only once at

the beginning. Then deflations are done with cheap rank-1 updates as $Q_{j+1} = Q_j - \frac{t_j t_j^\top}{t_j^\top t_j}$

Let

$$S_j = Q_j^\top Q_j, \quad M_j = S_j S_j,$$

and let a_j be the cSPCA loading vector solving

$$M_j a_j = S_j S_j a_j = \mu_j S a_j,$$

where $S = X^\top X$ is the original matrix and μ_j is the largest generalized eigenvalue. The score vector is always

$$t_j = X a_j,$$

so

$$t_j^\top t_j = a_j^\top S a_j.$$

With

$$Q_{j+1} = \left(I - \frac{t_j t_j^\top}{t_j^\top t_j} \right) Q_j,$$

define

$$g_j = Q_j^\top t_j, \quad d_j = t_j^\top t_j = a_j^\top S a_j.$$

Then

$$S_{j+1} = Q_{j+1}^\top Q_{j+1} = S_j - \frac{g_j g_j^\top}{d_j},$$

which is a rank-1 downdate. Consequently,

$$M_{j+1} = S_{j+1} S_{j+1} = M_j - \frac{S_j g_j g_j^\top + g_j g_j^\top S_j}{d_j} + \frac{g_j^\top g_j}{d_j^2} g_j g_j^\top.$$

Thus M_j is updated by a sum of rank-1 terms, avoiding recomputation of M_{j+1} from scratch.

Hence the update of M_j is a sum of rank-1 terms.

The generalized eigenvalue relation is consistent with these definitions, since

$$a_j^\top M_j a_j = a_j^\top S_j S_j a_j = \mu_j a_j^\top S a_j = \mu_j t_j^\top t_j,$$

where again μ_j is the largest generalized eigenvalue. This identity is useful for interpreting the scalar criterion associated with a_j , while the

actual deflation formulas above are expressed in terms of $g_j = Q_j^\top t_j$ and $d_j = t_j^\top t_j = a_j^\top S a_j$.

Using these rank-1 updates gives a substantial computational saving compared with recomputing the matrices from the deflated data matrix at each step. Recomputing the covariance matrix S_j from Q_j costs $O(np^2)$, whereas updating it directly costs only $O(p^2)$. For M_j , the saving is larger: recomputing it requires multiplying two $p \times p$ matrices, which costs $O(p^3)$, whereas the corresponding update uses only matrix-vector products and outer products, so its cost is $O(p^2)$. The reduction is therefore substantial, especially when p is large.

3. JSS EXAMPLE

4. Introduction: Count data regression in R

The introduction is in principle “as usual”. However, it should usually embed

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